Q1. Fetch the full name and hiring date of all Employees who work as Sales Representatives.

Ans:

Code -

```
SELECT
   CONCAT(firstname, ' ', lastname) AS full_name,
   hiredate
FROM employees
WHERE title = 'Sales Representative';
```

Output –

full_name	hiredate
Nancy Davolio	1992-05-01
Janet Leverling	1992-04-01
Margaret Peacock	1993-05-03
Michael Suyama	1993-10-17
Robert King	1994-01-02
Anne Dodsworth	1994-11-15

Insight - All 6 Sales Reps were hired between 1992–1994, meaning the entire sales team has 30+ years of tenure. The team has not expanded in a long time.

Recommendation - Consider hiring fresh sales talent to bring new energy and expand market reach. A growing business needs a growing sales team.

Q2. Which of the products in our inventory need to be reordered?

Ans:

Code –

```
SELECT productname, unitsinstock, unitsonorder, reorderlevel
FROM products
WHERE unitsinstock + unitsonorder <= reorderlevel
  AND discontinued = 0;
```

Output –

productname	unitsinstock	unitsonorder	reorderlevel
Nord-Ost Matjeshering	10	0	15
Outback Lager	15	10	30

Insight - Only 2 active products are below their reorder level. Nord-Ost Matjeshering has 0 units on order, making it the more urgent case right now.

Recommendation - Place an immediate reorder for Nord-Ost Matjeshering and confirm the Outback Lager order is on track. Set up automated reorder alerts to catch this earlier.

Q3. Find and display the details of customers who have placed more than 5 orders.

Ans:

Code –

```
SELECT c.customerid, c.companyname, c.contactname, COUNT(o.orderid) AS total_orders
FROM customers c
JOIN orders o ON c.customerid = o.customerid
GROUP BY c.customerid, c.companyname, c.contactname
HAVING COUNT(o.orderid) > 5
ORDER BY total_orders DESC;
```

Output –

customerid	companyname	contactname	total_orders
SAVEA	Save-a-lot Markets	Jose Pavarotti	31
ERNSH	Ernst Handel	Roland Mendel	30
QUICK	QUICK-Stop	Horst Kloss	28
HUNGO	Hungry Owl All-Night	Patricia McKenna	19
FOLKO	Folk och fä HB	Maria Larsson	19
BERGS	Berglunds snabbköp	Christina Berglund	18
HILAA	HILARION-Abastos	Carlos Hernández	18
RATTC	Rattlesnake Canyon	Paula Wilson	18
BONAP	Bon app'	Laurence Lebihan	17
LEHMS	Lehmanns Marktstand	Renate Messner	15

Insight - 63 customers have placed more than 5 orders. Top 3 — Save-a-lot, Ernst Handel, QUICK-Stop — are far ahead with 28–31 orders each. These are your VIP accounts.

Recommendation - Create a dedicated account manager for the top 10 customers. Offer them priority shipping and exclusive deals to lock in their loyalty long-term.

Q4. An employee (Margaret Peacock, EmployeeID 4) has the record of completing the most orders. However, there are some customers who've never placed an order with her. Show such customers.

Ans:

Code –

```
SELECT customerid, companyname, contactname
FROM customers
WHERE customerid NOT IN (
   SELECT DISTINCT customerid
   FROM orders
   WHERE employeeid = 4
);
```

Output –

customerid	companyname	contactname
CONSH	Consolidated Holdings	Elizabeth Brown
DUMON	Du monde entier	Janine Labrune
FISSA	FISSA Fabrica Inter. Salchichas S.A.	Diego Roel
FRANR	France restauration	Carine Schmitt
GROSR	GROSELLA-Restaurante	Manuel Pereira
LAUGB	Laughing Bacchus Wine Cellars	Yoshi Tannamuri
LAZYK	Lazy K Kountry Store	John Steel
NORTS	North/South	Simon Crowther
PARIS	Paris spécialités	Marie Bertrand
PERIC	Pericles Comidas clásicas	Guillermo Fernández
PRINI	Princesa Isabel Vinhos	Isabel de Castro
SANTG	Santé Gourmet	Jonas Bergulfsen
SEVES	Seven Seas Imports	Hari Kumar
SPECD	Spécialités du monde	Dominique Perrier
THEBI	The Big Cheese	Liz Nixon
VINET	Vins et alcools Chevalier	Paul Henriot

Insight - 16 customers have never interacted with Margaret — the best closer in the company. This is a clear missed opportunity that's sitting right in front of you.

Recommendation - Assign Margaret to reach out to these 16 customers. Her track record makes her the best person to convert or upsell these untouched accounts.

Q5. The developers at Cochin Traders are testing an app that the customers will use to show orders. In order to make sure that even the largest orders will show up correctly on the app, they'd like some samples of orders that have lots of individual line items. Display the top 10 orders with the most line items.

Ans:

Code –

```
SELECT orderid, COUNT(productid) AS line_item_count
FROM order_details
GROUP BY orderid
ORDER BY line_item_count DESC
LIMIT 10;
```

Output –

orderid	line_item_count
11077	25
10979	6
10847	6
10657	6
10337	5
10325	5
10294	5
10309	5
10324	5
10607	5

Insight - Order 11077 has 25 line items — nearly 4x more than the next largest order (6 items). This is a clear outlier and the perfect stress test case for the app.

Recommendation - Use Order 11077 as the primary stress test. Also test orders with 5–6 items as the "normal large order" baseline to ensure the app handles both gracefully.

Q6. Retrieve the top 5 best-selling products based on total quantity ordered.

Ans:

Code –

```
SELECT p.productname, SUM(od.quantity) AS total_quantity_sold
FROM order_details od
JOIN products p ON od.productid = p.productid
GROUP BY p.productname
ORDER BY total_quantity_sold DESC
LIMIT 5;
```

Output –

productname	total_qty_sold
Camembert Pierrot	1577
Raclette Courdavault	1496
Gorgonzola Telino	1397
Gnocchi di nonna Alice	1263
Pavlova	1158

Insight - All top 5 are food/cheese products, selling between 1,158–1,577 units. Camembert Pierrot leads by a solid margin. The product category is clearly a core strength.

Recommendation - Never let these 5 products go out of stock. Consider bundling them together as a "bestseller pack" to increase average order value.

Q7. Analyze the monthly order count for the year 1997.

Ans:

Code –

```
SELECT
   MONTHNAME(orderdate) AS month_name,
   MONTH(orderdate) AS month_num,
   COUNT(orderid) AS total_orders
FROM orders
WHERE YEAR(orderdate) = 1997
GROUP BY month_num, month_name
ORDER BY month_num;
```

Output –

month_num	month_name	total_orders
1	January	33
2	February	29
3	March	30
4	April	31
5	May	32
6	June	30
7	July	33
8	August	33
9	September	37
10	October	38
11	November	34
12	December	48

Insight - December 1997 had 48 orders — the highest of the year, 65% more than the slowest month (February with 29). Order volume clearly spikes in Q4.

Recommendation - Bulk up inventory and staff capacity before October every year. Run promotions in February and June (the two slowest months) to smoothen the demand curve.

Q8. Calculate the difference in sales revenue for each month compared to the previous month.

Ans:

Code –

```
WITH monthly_revenue AS (
   SELECT
      DATE_FORMAT(o.orderdate, '%Y-%m') AS month,
      ROUND(SUM(od.unitprice * od.quantity * (1 - od.discount)), 2) AS revenue
   FROM orders o
   JOIN order_details od ON o.orderid = od.orderid
   GROUP BY month
)
SELECT
   month,
   revenue,
   ROUND(revenue - LAG(revenue) OVER (ORDER BY month), 2) AS revenue_diff
FROM monthly_revenue
ORDER BY month;
```

Output –

month	revenue	revenue_diff
1996-07	27861.90	–
1996-08	25485.28	-2376.62
1996-10	37515.72	+11134.32
1997-01	61258.07	+16018.44
1997-02	38483.64	-22774.43
1997-06	36362.80	-17418.49
1997-12	71398.43	+27864.62
1998-01	94222.11	+22823.68
1998-04	123798.68	+18944.52
1998-05	18333.63	-105465.05

Insight – Revenue grew strongly through 1998 but two big drops stand out, Feb 1997 (−$22K) and Jun 1997 (−$17K). The May 1998 drop is just incomplete data, not a real crash.

Recommendation - Investigate what caused Feb and Jun 1997 drops, was it fewer orders or lower value orders? Use that insight to build a mid-year sales push to prevent repeat dips.

Q9. Calculate the percentage of total sales revenue for each product.

Ans:

Code –

```
SELECT
   p.productname,
   ROUND(SUM(od.unitprice * od.quantity * (1 - od.discount)), 2) AS product_revenue,
   ROUND(
      SUM(od.unitprice * od.quantity * (1 - od.discount)) * 100.0 /
      SUM(SUM(od.unitprice * od.quantity * (1 - od.discount))) OVER (), 2
   ) AS revenue_pct
FROM order_details od
JOIN products p ON od.productid = p.productid
GROUP BY p.productname
ORDER BY revenue_pct DESC;
```

Output –

productname	product_revenue	revenue_pct
Côte de Blaye	141396.73	11.17%
Thüringer Rostbratwurst	80368.67	6.35%
Raclette Courdavault	71155.70	5.62%
Tarte au sucre	47234.97	3.73%
Camembert Pierrot	46825.48	3.70%
Gnocchi di nonna Alice	42593.06	3.36%
Manjimup Dried Apples	41819.65	3.30%
Alice Mutton	32698.38	2.58%
Carnarvon Tigers	29171.88	2.30%
Rössle Sauerkraut	25696.64	2.03%

Insight - Côte de Blaye alone contributes 11.17% of all revenue — nearly double the #2 product. The top 3 products together account for ~23% of total business revenue.

Recommendation - Treat Côte de Blaye as a flagship product — it must always be in stock. Review the bottom 40 products (likely contributing under 0.5% each) for potential discontinuation.

Q10. Determine the cumulative percentage of total sales revenue for each month.

Ans:

Code –

```
WITH monthly_revenue AS (
   SELECT
      DATE_FORMAT(o.orderdate, '%Y-%m') AS month,
      ROUND(SUM(od.unitprice * od.quantity * (1 - od.discount)), 2) AS revenue
   FROM orders o
   JOIN order_details od ON o.orderid = od.orderid
   GROUP BY month
)
SELECT
   month,
   revenue,
   ROUND(
      SUM(revenue) OVER (ORDER BY month) * 100.0 / SUM(revenue) OVER (), 2
   ) AS cumulative_pct
FROM monthly_revenue
ORDER BY month;
```

Output –

month	revenue	cumul_revenue	cumul_pct
1996-07	27861.90	27861.90	2.20%
1996-12	45239.63	208083.97	16.44%
1997-06	36362.80	489549.94	38.68%
1997-09	55629.24	643487.71	50.84%
1997-12	71398.43	825169.18	65.19%
1998-02	99415.29	1018806.58	80.49%
1998-04	123798.68	1247459.42	98.55%
1998-05	18333.63	1265793.05	100.00%

Insight - 50% of all-time revenue wasn't reached until September 1997, and 80% only by February 1998. Revenue is heavily back-loaded, accelerating sharply in 1998.

Recommendation - Plan cash flow carefully for the early months (low cumulative %). The strong 1998 growth trend suggests the business is scaling — invest in supply chain and team capacity now to sustain it.